Research Evolution — Joby Aviation (JOBY), FIN 43900 Project 1

Tobey Uhm · Edition A baseline of record: Brightspace "Lab 03 — Project 1 Valuation Edition A", attempt 1 of 3, written Sept. 1, 2026, 2:30–2:31 PM ET (receipt transcription in the evidence package). The local Edition A file was created Sept. 1, 2026, 2:30:21 PM ET; SHA-256 `CDF3C3529352D722F0116822F4522BBF9BDE875B1E719A861A1A41465C6B6F99`. It is reproduced below **unchanged**.

Student attestation (recorded Oct. 19, 2026): I attest that Edition A was written without generative-AI assistance before the Brightspace receipt. — Tobey Uhm

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Edition A — unchanged Brightspace baseline (verbatim)

Project 1 — Edition A

Decision, user, company, and valuation date

Decision user: A buy-side investment committee with no current position in Joby Aviation, Inc. (NYSE: `JOBY`).

Decision: Whether to initiate research coverage and advance Joby to a full FCFF/enterprise-value valuation, rather than whether to purchase the stock immediately.

Company: Joby Aviation, Inc. (`JOBY`; CIK 0001819848), an electric vertical takeoff and landing (eVTOL) aircraft developer and emerging air-mobility operator.

Valuation date: September 1, 2026. Financial statement reference period: fiscal year ended December 31, 2025.

Initial thesis — watch / defer

Watch / defer; do not initiate a position yet. Joby is admissible for deeper research because it is an operating company with a meaningful reported service-revenue base and a large liquid balance-sheet resource relative to its present operating scale. However, a purchase decision is premature: 2025 revenue was driven by acquired and ancillary services while the core eVTOL business remains dependent on certification, commercial launch, manufacturing execution, customer adoption, and continued financing. The valuation will be highly sensitive to the date and scale of durable commercial cash flows, not just a near-term revenue multiple.

Known evidence — primary filing

Primary filing: Joby Aviation, Inc., Form 10-K, accession number `0001819848-26-000160`, filed February 27, 2026, for the fiscal year ended December 31, 2025. The financial statements are presented in U.S. dollars and **in thousands**, except share and per-share amounts.

Evidence item	Filing-supported fact	Why it matters
Entity	Joby Aviation, Inc.	Legal issuer and operating-company target.
Cash	$240.810 million cash and cash equivalents at December 31, 2025	Immediate liquidity available to fund development and operations.
Short-term investments	$1.167 billion	Included with cash in assessing liquidity; should not be confused with operating revenue.
Liquid resources	$1.408 billion cash, equivalents, and short-term investments	Establishes the initial funding runway context.
Debt	No short- or long-term funded-debt line was reported on the December 31, 2025 balance sheet. The filing subsequently reports $690.0 million principal of 0.75% convertible senior notes issued February 2, 2026.	Enterprise-value bridge must distinguish the balance-sheet date from the post-year-end capital structure.
Diluted-share evidence	826,240,955 weighted-average basic **and diluted** shares for 2025; the filing says the amounts are identical because net loss made potential dilution antidilutive. 979,334,237 common shares were outstanding on February 23, 2026.	A valuation must state whether it uses weighted-average diluted shares or period-end shares, and reconcile post-year-end financing separately.
2025 revenue	$53.425 million	Initial evidence of an operating revenue base, but not evidence that eVTOL service economics are mature.

Evidence item	Filing-supported fact	Why it matters
Operating-value driver	Revenue grew from $0.136 million in 2024 to $53.425 million in 2025, primarily from Blade passenger-service revenue after the acquisition plus DOD/on-base operations, demonstration flights, and third-party engineering services.	Near-term value depends on conversion from these services and development activity to scalable commercial air-taxi operations.
Development cost	$581.101 million of R&D expense in 2025	Highlights the capital intensity and the importance of certification/manufacturing timing.

Source locations in the 10-K

- Consolidated balance sheets and statements of operations: pages 52–53 of the filing (cash, investments, liabilities, revenue, R&D, and weighted-average shares).
- Liquidity discussion: page 42 (cash/investments and funding sources).
- Net loss per share: Note 14, page 88 (why diluted shares equal basic shares).
- Subsequent events: page 90 (February 2026 equity offering and $690.0 million 2032 convertible notes).

Synthetic enterprise-to-equity bridge check

Source: Week 2 Offline/No-Code Fallback — Valuation Object and Bridge. This is a **SYNTHETIC TEACHING CASE**, not JOBY market or filing evidence. All monetary amounts are USD millions and diluted shares are millions.

Step	Amount	Check
Enterprise value	$1,000.0	Operating-business value.
Plus: non-operating cash	$100.0	Non-operating asset available to common equity after claims.
Less: debt	$(250.0)	Debt claim.
Less: noncontrolling interest	$(20.0)	Non-common claim.
Less: unfunded pension	$(10.0)	Debt-like claim.
Equals: synthetic common equity value	**$820.0**	Matches the class reconciliation anchor.
Divided by: diluted shares	100.0	Million shares; consistent with USD millions.
Equals: implied common equity value per share	**$8.20**	Matches the class reconciliation anchor.

Companion synthetic FCFF check: NOPAT = `EBIT $150.0 × (1 − 25%) = $112.5`; FCFF = `$112.5 + D&A $40.0 − capex $60.0 − increase in operating NWC $15.0 = $77.5`. These results match the class reconciliation anchors.

Unit/sign failure to avoid: subtracting cash rather than adding it in the enterprise-to-equity bridge would produce $620.0 million of equity value, or $6.20 per share, and reverse the interpretation by $2.00 per share. For JOBY, a later bridge must also explicitly address the February 2026 convertible notes and equity issuance rather than mechanically using the December 31 balance sheet.

Consequential assumptions

1. Joby can obtain the necessary certification and operating approvals on a timetable that permits commercial service before liquidity is materially impaired.
2. The company can manufacture aircraft and develop operating infrastructure at a cost and pace consistent with an eventual positive FCFF profile.
3. Passenger demand, route economics, utilization, pricing, and safety acceptance can support returns above the company's cost of capital after launch.
4. Future capital raises, convertible-note conversion, warrants, earnouts, and equity compensation will not dilute per-share value beyond the economics assumed in a full model.
5. Cash and short-term investments are available as non-operating value in the bridge only to the extent they are not required to fund the operating plan.

Unknowns and risks to resolve

- The certified commercial-launch date, market-by-market regulatory permissions, and probability of delay.
- The normalized revenue, margin, capital-expenditure, and working-capital profile of a scaled eVTOL service.

- Aircraft production rate, supply-chain reliability, and cost per aircraft.
- The portion of reported 2025 revenue and assets that is recurring and representative of the intended eVTOL operating model.
- The appropriate treatment and valuation impact of the 2032 convertible notes, warrants, earnout liabilities, lease liabilities, and other potentially dilutive securities.
- The required return, terminal assumptions, and scenario probabilities appropriate for an early-commercialization aerospace business.

Research and validation plan

1. Build a dated capital-structure schedule from the 2025 10-K and subsequent 2026 SEC filings, separating period-end common shares, weighted-average diluted shares, options/RSUs/warrants, earnouts, and the 2032 convertible notes.
2. Reconcile cash, investments, debt, and debt-like claims to a single valuation date before calculating equity value.
3. Build base, delayed-certification, and accelerated-execution FCFF scenarios. Make launch timing, aircraft deliveries, utilization, price, contribution margin, R&D, and capital spending explicit drivers.
4. Validate certification and launch milestones against FAA/regulatory documentation and the company's SEC disclosures, not against a search-result summary.
5. Separate acquired service revenue from the forecast economics of Joby's intended eVTOL network; use a comparable-company and precedent-transaction framework only as a cross-check.
6. Run sensitivity tables for launch delay, production rate, utilization, fare/yield, margin, dilution, and discount rate; identify the inputs that move equity value per share most.
7. Complete the synthetic enterprise-to-equity bridge check using the class inputs and preserve the result alongside the model.

Human-partner falsification question

Question (no answer proposed): "What evidence would show that Joby cannot turn FAA certification and early flight operations into a commercially scalable, cash-generative air-taxi business before additional financing or dilution becomes necessary?"

Before checkpoint submission, have a human learning partner ask or explicitly adopt this question and record that partner in the Lab 03 checkout. No answer is included here, as required.

Submission-control note

Before submitting this as the Edition A checkpoint, have the stated falsification question asked or explicitly adopted by a human learning partner. Then submit the file once, preserve that exact submitted version unchanged, and record the checkpoint receipt identifier in the lab checkout.

Dated correction addendum

None required. Edition A's factual statements were rechecked on Oct. 19, 2026. EDGAR shows the FY2025 10-K filed **Feb. 27, 2026**, as Edition A states; the "Feb. 26" date in the later Lab 06/Lab 11 code was the error, and Edition B uses Feb. 27. Later facts (the convertible notes, the 2026 equity offering, the June 30, 2026 balance sheet) are **updates**, not corrections, so they go in Edition B below.

Edition B — after project-specific research and disclosed AI assistance

Date: October 19, 2026 · **Valuation date:** September 1, 2026 · **Repository:** `Joby-research` · **Frozen output:** `output/joby_valuation_report.md`

Current decision and intended user

The intended user is unchanged: **a buy-side investment committee with no current JOBY position.** The decision has moved from Edition A's narrower question ("should Joby advance to a full FCFF valuation?") to the full Project 1 decision: **initiate, watch/defer, or do not initiate**.

Edition B position: Watch / defer — no position at $6.69. The integrated FCFF/WACC valuation gives **$0.00–$3.82 per share** (base $0.34). The reverse DCF shows the price needs about **2.1×** our base fleet in every year. A two-year certification delay takes modeled equity value below zero, which would mean **Do not initiate**. To initiate, we need an FAA type certificate **and** a price at or below 75% of the updated base value.

A-to-B evidence table

Changed item	Edition A (Sept. 1, 2026)	Edition B (Oct. 19, 2026)	Primary driver of change	Evidence or test that justified the change
Thesis / recommendation	Watch/defer; no position yet; valuation not yet built	**No material change in action** (watch/defer, no position), now with a $0.00–$3.82 range and explicit initiate / do-not-initiate rules	Own analysis, with AI-assisted model construction (mixed)	Base $0.34 and upside $3.82 are both below $6.69 (`output/joby_valuation_report.md`, Scenarios). Reverse DCF needs 2.13× fleet. No WACC × g cell reaches $6.69.
Data / source choice	FY2025 10-K, Dec. 31, 2025 balance sheet; 826.2M weighted-average diluted shares	**Q2 2026 10-Q** (latest filing before Sept. 1): June 30, 2026 balance sheet, 989.1M shares on Aug. 3, 2026, antidilution table, convertible-note terms; Sept. 1 close of $6.69	Course instruction (use the latest authoritative filing at the valuation date) and own analysis	Test `test_h1_actuals_reconcile_to_10q_operating_loss` (−$494.463M); diluted-share test 1,051.26M; source ledger S4.
Assumption / model design	Planned FCFF scenarios with launch timing, deliveries, utilization and price as explicit drivers	Integrated IS/BS/CF → FCFF → WACC → bridge. Revenue = fleet × flight hours × revenue per flight hour + Blade services. Lab 11's top-down $3.5B 2030 revenue replaced by a driver build ($1.13B base). FCFE replaced by FCFF.	**Peer challenge** (Eric Boyd, Labs 11–12: "What primary evidence supports a $75M/…/$3.5B revenue path…?" and "find stronger evidence for … fleet growth, fares, and utilization"), plus own analysis and AI-assisted coding	FY2026 guidance of $115–125M vs Lab 11's $75M; 12 aircraft in production (S7). Balance check passes every period; FCFF identity test; Week 3 known answer reproduced through the same discounting function.
Risk, limitation, or uncertainty	Certification timing, dilution, convertible-note treatment and the cash-as-runway	Quantified: base needs about **$5.0B** of new equity (first raise 2027); delay case needs $12.7B and gives $0 value;	Mixed: own analysis + course instruction (Week 6 downside through linked statements) + AI	Scenario table; tests `test_convertible_treated_as_debt_below_conversion_price` and `test_cash_never_below_floor`.

Changed item	Edition A (Sept. 1, 2026)	Edition B (Oct. 19, 2026)	Primary driver of change	Evidence or test that justified the change
	question listed but not quantified	about 22 months of runway at the FY2027 burn. Convertible treated as debt because the $14.19 conversion price exceeds value. The locked changed-input test (Oct. 19) showed revenue per flight hour is decision-critical: $2,000 instead of $2,500 gives −$1.34/share → Do not initiate; breakeven ≈ $2,390.		
Code reliability	—	Three Lab 11 defects repaired: (1) a four-year path silently accepted; (2) a different discount rate used for the terminal value; (3) g ≥ r accepted	AI (an AI review flagged them), verified by own tests	`RepairedDefects` tests fail on the original logic and pass on the repair; Lab 11 results ($375.1M / $121.2M) unchanged.

What changed from Edition A to Edition B, and why?

The action did not change. The evidence behind it did. Edition A said watch/defer because the core eVTOL business was unproven and the valuation was still to be built. Edition B builds that valuation and finds the same answer for a measurable reason: on a driver-based ramp, the PV of FCFF from 2026 H2 to 2040 (about −$3.4B, driven by burn through 2034) outweighs the PV of the terminal value (+$2.3B), so equity value rests almost entirely on today's $2.26B of cash. At $6.69 the market is paying for about twice the fleet scale our evidence supports.

The most consequential **model** change came from **peer challenge**. Eric Boyd asked what primary evidence supported Lab 11's $3.5B 2030 revenue. We had no answer, so the top-down path was replaced with fleet × utilization × price, anchored to FY2026 guidance and disclosed production. The **data** change (the June 30, 2026 balance sheet, the post-convertible capital structure and the Aug. 3 share count) followed the **course** rule to use the latest filing at the valuation date. **AI** contributed coding speed, the defect review and filing extraction. Every AI-supplied number or claim was checked against the SEC text or a test before it was used (see `Validation-and-AI-Use.pdf`). The unchanged action would flip to **Do not initiate** on a certification slip past Dec. 31, 2027. It would move toward **Initiate** only with a type certificate plus a price at or below 75% of the base value.